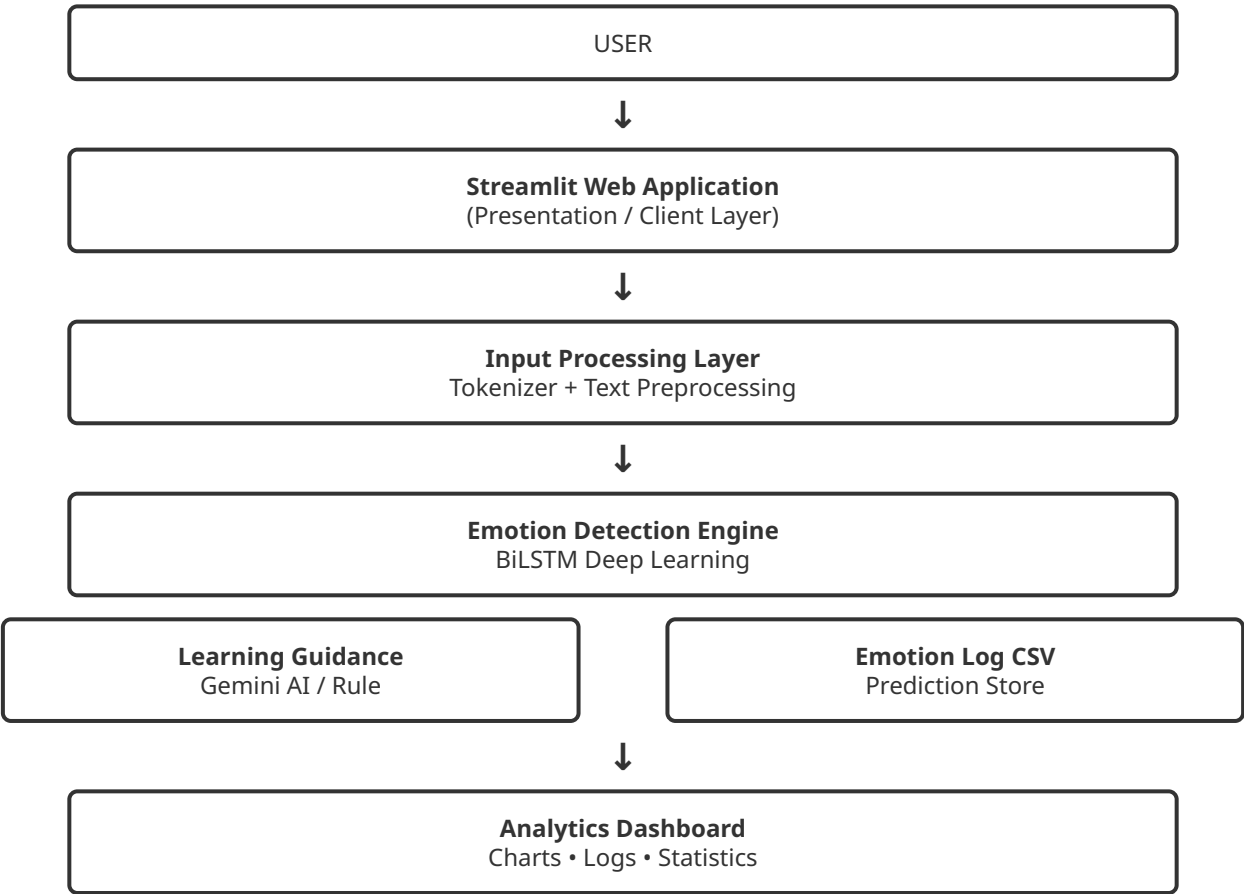


Solution Architecture

Date: 14 July 2026 | **Team ID:** | **Project Name:** Emotion Detection and Learning Support Engine

The Solution Architecture illustrates the complete workflow of the Emotion Detection and Learning Support Engine. The system accepts user text through a Streamlit web interface, preprocesses the input using NLP techniques, predicts emotions using a trained BiLSTM model, generates personalized learning guidance, stores prediction logs, and presents analytics through an interactive dashboard.



Component Name	Description / Role	Technology Used
Presentation Layer	Provides the user interface for entering emotions and displaying predictions, guidance, and analytics.	Streamlit
Input Processing Layer	Cleans and converts user text into token sequences suitable for model prediction.	Python, Tokenizer, Keras Preprocessing
Emotion Detection Engine	Predicts user emotions using a trained deep learning model.	TensorFlow, Keras, BiLSTM
Learning Guidance Module	Generates personalized learning suggestions based on detected emotions.	Gemini API / Rule-based Guidance
Emotion Log Storage	Stores prediction history, timestamps, and confidence scores.	CSV Files, Pandas
Analytics Dashboard	Displays emotion distribution, recent prediction logs, and visual insights.	Matplotlib, Pandas, Streamlit
Model Files	Stores the trained model, tokenizer, and label encoder used for inference.	.h5 Model, Pickle (.pkl) Files